1. A body cools in 7 minutes from $60^{\circ}C$ to $40^{\circ}C$. The temperature of the surrounding is $10^{\circ}C$. The temperature of the body after the next 7 minutes will be

[2023 (06 Apr Shift 2)]

(1) $30^{\circ}C$
(2) $32^{\circ}C$
(3) $34^{\circ}C$
(4) $28^{\circ}C$

2. A steel rod of length 1 m and cross-sectional area $10^{-4}\ m^2$ is heated from $0^{\circ}C$ to $200^{\circ}C$ without being allowed to extend or bend. The compressive tension produced in the rod is _____ $\times 10^4$ N. (Given Young's modulus of steel $= 2 \times 10^{11}\ N\ m^{-2}$, coefficient of linear expansion $= 10^{-5}\ K^{-1}$)

[2023 (08 Apr Shift 2)]

3. 1 kg of water at $100^{\circ}C$ is converted into steam at $100^{\circ}C$ by boiling at atmospheric pressure. The volume of water changes from $1.00 \times 10^{-3}\ m^3$ as a liquid to $1.671\ m^3$ as steam. The change in internal energy of the system during the process will be (Given latent heat of vaporisation $= 2257$ kJ / kg, Atmospheric pressure $= 1 \times 10^5$ Pa)

[2023 (11 Apr Shift 1)]

(1) -2426 kJ
(2) $+2090$ kJ
(3) -2090 kJ
(4) $+2476$ kJ

4. On a temperature scale 'X', the boiling point of water is $65^{\circ}X$ and the freezing point is $-15^{\circ}X$. Assuming that the X scale is linear. The equivalent temperature corresponding to $-95^{\circ}X$ on the Fahrenheit scale would be

[2023 (11 Apr Shift 1)]

(1) $-112^{\circ}F$
(2) $-48^{\circ}F$
(3) $-148^{\circ}F$
(4) $-63^{\circ}F$

5. A body cools from $80^{\circ}C$ to 60° in 5 minutes. The temperature for the surrounding is $20^{\circ}C$. The time it takes to cool from $60^{\circ}C$ to $40^{\circ}C$ is

[2023 (12 Apr Shift 1)]

(1) 450 s
(2) 420 s
(3) 500 s
(4) $\frac{25}{3}$ s

6. Two plates A and B have thermal conductivities $84\ W\ m^{-1}\ K^{-1}$ and $126\ W\ m^{-1}\ K^{-1}$ respectively. They have same surface area and same thickness. They are placed in contact along their surfaces. If the temperatures of the outer surfaces of A and B are kept at $100^{\circ}C$ and $0^{\circ}C$ respectively, then the temperature of the surface of contact in steady state is_____ $^{\circ}C$.

[2023 (13 Apr Shift 2)]

ANSWER KEYS

1. (4) **2.** (4) **3.** (2) **4.** (3) **5.** (3) **6.** (40)

1. (4)

Newton's law of cooling states that

$\frac{\left(T_1-T_2\right)}{t}=K\left(\left(\frac{T_1+T_2}{2}\right)-T_0\right) \quad \ldots(i)$

where T_0 is the temperature of the surrounding. From equation (i),

$\frac{60-40}{7 \times 60}=K\left(\frac{60+40}{2}-10\right)$

$\Rightarrow \frac{20}{7 \times 60}=40 K$

$\Rightarrow K=\frac{20}{7 \times 60 \times 40}$

Let the temperature of the body after the given time be $x\ ^{\circ}\mathrm{C}$. By Newton's law of cooling,

$\frac{40-x}{7 \times 60}=K\left(\frac{40+x}{2}-10\right)$

$\Rightarrow \frac{40-x}{7 \times 60}=\frac{1}{14 \times 60}\left(\frac{20+x}{2}\right)$

$\Rightarrow 160-4 x=20+x$

$\Rightarrow x=28\ ^{\circ}\mathrm{C}$

2. (4)

The formula to calculate the Young's modulus of the material of the wire is given by

$Y=\frac{F L}{A \Delta L} \quad \ldots(1)$

The increase in length of the wire due to increase in temperature is given by

$\Delta L=L \alpha\left(T_2-T_1\right) \quad \ldots(2)$

Substitute the expression for the extension in length from equation (2) into equation (1) and simplify to obtain the required compressive tension.

$Y=\frac{F L}{A L \alpha\left(T_2-T_1\right)}$

$\Rightarrow F=A \alpha Y\left(T_2-T_1\right) \quad \ldots(3)$

Substitute the values of the known parameters into equation (3) to calculate the required compressive tension in the wire.

$F=10^{-4}\ m^2 \times 10^{-5}\ \mathrm{K}^{-1} \times 2 \times 10^{11}\ \mathrm{Nm}^{-2} \times\left(200^{\circ}\ \mathrm{C}-0^{\circ}\ \mathrm{C}\right)$

$=4 \times 10^4\ \mathrm{N}$

3. (2)

The work to be done in the process is given by

$dW=PdV$

$=1 \times 10^5\ \mathrm{Pa} \times(1.671-0.001)\ \mathrm{m}^3$

$=1.670 \times 10^5\ \mathrm{J}$

The change in heat energy during the vaporisation process can be calculated as follows-

$\Delta Q_{\text{supplied}}=2257 \times 1 \times 10^3\ \mathrm{J}$

$=22.57 \times 10^5\ \mathrm{J}$

Hence, the change in internal energy in the process is given by

$\Delta U=\Delta Q_{\text{supplied}}-\Delta W$

$=(22.57-1.67) \times 10^5\ \mathrm{J}$

$=20.9 \times 10^5\ \mathrm{J}$

$=2090\ \mathrm{kJ}$

4. (3)

The relation between two temperature scales can be written as follows

$\frac{65-x}{65-(-15)}=\frac{212-F}{180} \quad \ldots(1)$

Substitute the values of the known parameters in to equation (1) and sole to calculate the required temperature in Fahrenheit scale.

$\frac{65+95}{80}=\frac{212-F}{180}$

$\Rightarrow 2 \times 180=212-F$

$\Rightarrow F=212-360$

$=-148^{\circ}\mathrm{F}$

5. (3)

The formula to calculate the rate of cooling of the object is given by

$\frac{dT}{dt} = K\left[\frac{T_f+T_i}{2} - T_0\right] \quad \ldots(1)$

For the first case, it can be written, using equation (1) that

$\frac{80-60}{5} = K\left[\frac{80+60}{2} - 20\right]$

$4 = 50K \quad \ldots(2)$

If t is the required time for the second case, from equation (1), it can be written that

$\frac{60-40}{t} = K\left[\frac{60+40}{2} - 20\right]$

$\frac{20}{t} = 30K \quad \ldots(3)$

Divide equation (2) by equation (3) and solve to calculate the required time.

$\frac{4}{\frac{20}{t}} = \frac{50K}{30K}$

$\Rightarrow \frac{t}{5} = \frac{5}{3}$

$\Rightarrow t = \frac{25}{3} \text{ min} = 500 \text{ s}$

6. (40)

Let the temperature of the contact surface be T,

So, $H_A = H_B$

$\frac{K_A A'(T_A - T)}{L} = \frac{K_B A'(T - T_B)}{L}$

$\Rightarrow 84\left(100 - T\right) = 126\left(T - 0\right)$

$\Rightarrow 2\left(100 - T\right) = 3T$

$\Rightarrow T = 40\ ^\circ\text{C}$